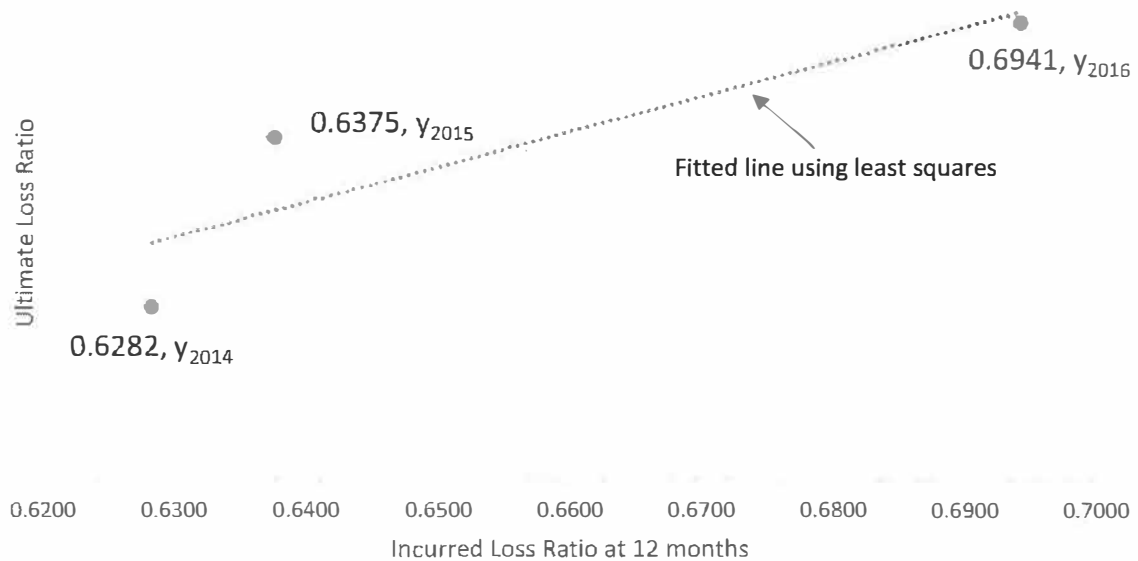


4. (2 points)

Given the following information:



- The equation for the line in the graph above is $y = 0.1126 + bx$.
- $\bar{xy} = 0.5979$
- Cumulative incurred loss for accident year 2017 at 12 months is \$6,000,000.
- Earned premium for accident year 2017 is \$8,000,000.
- Assume there is no further development after 24 months.

Calculate the ultimate loss for accident year 2017 using the least squares method.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 4	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE(S): A1
SAMPLE ANSWERS	
<u>Sample 1</u> $\bar{x} = \frac{0.6282 + 0.6375 + 0.6941}{3} = 0.6533$ $\bar{x}^2 = 0.4268$ $\overline{x^2} = \frac{0.6282^2 + 0.6375^2 + 0.6941^2}{3} = 0.4276$ $\bar{y} = 0.1126 + 0.6533b$ $b = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2} = \frac{0.5979 - 0.6533\bar{y}}{0.4276 - 0.4268}$ $b = \frac{0.5979 - 0.6533 \times (0.1126 + 0.6533b)}{0.0008} = 1.2261$ For AY 2017, $y = 0.1126 + 1.2261x = 0.1126 + 1.2261 \times \frac{6,000,000}{8,000,000} = 1.0322$ $Ultimate = 1.0322 \times 8,000,000 = 8,257,600$ <u>Sample 2</u> $\overline{XY} = \overline{XY} = 0.5979 = \frac{1}{3} \times (0.6282 \times \hat{Y}_1 + 0.6375 \times \hat{Y}_2 + 0.6941 \times \hat{Y}_3)$ $= \frac{1}{3} \times [0.6282 \times (a + 0.6282b) + 0.6375 \times (a + 0.6375b) + 0.6941 \times (a + 0.6941b)]$ $= \frac{1}{3} \times (1.9598a + 1.2828b)$ $= \frac{1}{3} \times (1.9598 \times 0.1126 + 1.2828b)$ $b = 1.2261$ $Ult_{2017} = \left(0.1126 + 1.2261 \times \frac{6,000,000}{8,000,000}\right) \times 8,000,000 = 8,257,600$	
EXAMINER'S REPORT	
Candidates were expected to compute the least square estimate by applying and solving the equations for coefficients a and b. Common mistakes included: • Not calculating AY 2017 projected ultimate losses. • Deriving AY 2017 Projected ultimate loss as a product of AY 2017 projected ultimate loss	

EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

ratio and AY 2017 incurred loss.

- Estimating $\bar{Y}$ from the graph